DUALIZING COMPLEX AND SHEAF OF A CLOSED SUBSCHEME BY RSHEAFHOM AGAINST THE AMBIENT ONES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the adjoint composition, the RSheafHom identification, the H^{-1} extraction, and the pushforward upgrade are all retained verbatim.

Editorial note. The label lemma-closed-immersion-dim-1-CM names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Twisted inverse images as right adjoints. Duality for Schemes, Lemma duality-lemma-twisted-inverse-image and duality-lemma-twisted-inverse-image-closed. Provides the adjoints a, b, c of f, g, i, the composition b = c composed with a from uniqueness of right adjoints, and the RSheafHom description of c.
- The curve duality framework. Same chapter, Lemma lemma-duality-dim-1. Fixes omega_X^bullet = a(O_Spec(k)) and omega_X = H^{-1}, the objects related by the lemma.
- Sheaf Hom against ideals. Duality for Schemes, Lemma duality-lemma-sheaf-with-exact-support-ext. Upgrades the H^{-1} identification to i_*omega_Y = SheafHom_{O_X}(i_*O_Y, omega_X).

Source reference. Original lemma 0A77: sources/stacks-chow/tags.txt, source lines 12870–12870. Prerequisite pointer invoked at proof line(s) 1008; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A9E: sources/stacks-chow/tags.txt, source lines 12947–12947. Prerequisite pointer invoked at proof line(s) 992; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A9X: sources/stacks-chow/tags.txt, source lines 12965–12965. Prerequisite pointer invoked at proof line(s) 1001; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS2: sources/stacks-curves/pinned-original-source.tex, source lines 796–796. Prerequisite pointer invoked at proof line(s) 980;

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it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a proper scheme of dimension ≤ 1 over a field k. Let $i : Y \to X$ be a closed immersion. Let $\omega_X^\bullet$, ω_X, $\omega_Y^\bullet$, ω_Y be as in Lemma 0BS2. Then*

(1) $\omega_Y^\bullet = R\mathcal{H}om(\mathcal{O}_Y, \omega_X^\bullet)$,
(2) $\omega_Y = \mathcal{H}om(\mathcal{O}_Y, \omega_X)$ *and* $i_*\omega_Y = \mathcal{H}om_{\mathcal{O}_X}(i_*\mathcal{O}_Y, \omega_X)$.

Proof. Denote $g : Y \to \mathrm{Spec}(k)$ and $f : X \to \mathrm{Spec}(k)$ the structure morphisms. Then $g = f \circ i$. Denote a, b, c the right adjoint of Duality for Schemes, Lemma 0A9E for f, g, i. Then $b = c \circ a$ by uniqueness of right adjoints and because $Rg_* = Rf_* \circ Ri_*$. In the proof of Lemma 0BS2 we set $\omega_X^\bullet = a(\mathcal{O}_{\mathrm{Spec}(k)})$ and $\omega_Y^\bullet = b(\mathcal{O}_{\mathrm{Spec}(k)})$. Hence $\omega_Y^\bullet = c(\omega_X^\bullet)$ which implies (1) by Duality for Schemes, Lemma 0A9X. Since $\omega_X = H^{-1}(\omega_X^\bullet)$ and $\omega_Y = H^{-1}(\omega_Y^\bullet)$ we conclude that $\omega_Y = \mathcal{H}om(\mathcal{O}_Y, \omega_X)$. This implies $i_*\omega_Y = \mathcal{H}om_{\mathcal{O}_X}(i_*\mathcal{O}_Y, \omega_X)$ by Duality for Schemes, Lemma 0A77. □